

Team Members

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Project Title

A Critical Analysis of AI Coding Agents in Open Source Development

Motivation

AI coding assistants such as [GitHub Copilot](#), [Cursor](#), [Claude Code](#), [Gemini](#), and [ChatGPT](#) are becoming more common in software development. Many developers now use these tools to help write code, generate pull requests, fix bugs, and improve productivity. Because of this, it is important to study how AI-assisted contributions behave in real open-source software projects.

This project focuses on understanding whether pull requests associated with different AI coding agents show differences in merge success, review patterns, and repository acceptance behavior. The project is connected to the broader area of repository mining and software engineering analytics, where researchers analyze software repositories to better understand development practices.

The project is also motivated by the growing interest in AI-assisted software engineering research, especially in the Mining Software Repositories (MSR) community. While there has already been research on AI-assisted development, there is still limited comparative analysis between different AI coding agents and how repositories respond to them.

This study combines software engineering concepts with data science techniques such as exploratory analysis, visualization, statistical testing, and machine learning. The project is designed to be realistic for a semester-long course project while still exploring meaningful research questions.

Research Questions

The project will investigate the following research questions:

1. Which AI coding agents appear most frequently in the dataset?
2. Do pull requests associated with different AI coding agents have different merge or approval rates?

3. Which AI coding agents produce pull requests that are merged the fastest?
4. How do pull request metadata features affect merge success and review duration?
5. Are certain metadata patterns associated with better or worse outcomes for specific AI coding agents?
6. Can machine learning models predict pull request merge success or review outcomes using pull request metadata?
7. What limitations and threats to validity exist in the dataset and methodology, and how might they affect the conclusions?

Expected Data Sources

The main dataset for this project will be the AIDev dataset from the MSR Challenge:

[AIDev Dataset \(MSR Challenge\)](#)

The dataset contains AI-related pull request activity, repository metadata, commit information, and development workflow data collected from GitHub repositories. It is designed for studying AI-assisted software engineering activity and is well suited for this project.

The dataset includes information related to:

- Pull requests
- Repository activity
- Commits
- Review activity
- AI-related development traces
- Metadata associated with development workflows

Additional metadata may optionally be collected using:

- [GitHub REST API](#)
- [GitHub GraphQL API](#)

These APIs may be used to retrieve:

- Additional pull request details
- Updated repository statistics
- Review comments
- Repository activity information
- Missing metadata fields

Potential metadata features used in the analysis include:

- Pull request title length
- Pull request body length
- Number of commits
- Number of changed files
- Additions and deletions
- Labels/tags

- Review comments
- Merge status
- Time to merge
- Repository popularity/activity
- Mentions of AI tools in commits or pull request descriptions

Preliminary Dataset Exploration

The AIDev dataset was successfully accessed and explored using Python and Jupyter Notebook. Initial pilot experiments were performed to confirm that the dataset is accessible and suitable for the proposed research questions.

During the initial exploration phase, the dataset was loaded using pandas and inspected to better understand:

- Dataset structure
- Available metadata fields
- Pull request information
- AI-related attributes
- Repository activity information

Basic filtering and inspection experiments were also performed to confirm that the dataset contains enough information for:

- Merge outcome analysis
- Review pattern analysis
- Descriptive analytics
- Predictive modeling

The pilot exploration confirmed that the dataset is appropriate for a semester-long project and supports the planned analysis workflow.

Research Strategies / Methodology

Data Collection

The project will primarily use the AIDev dataset as the main source of data. The dataset will be downloaded and processed using Python-based tools in Jupyter Notebook.

Additional metadata may optionally be collected through the GitHub REST API and GraphQL API if certain repository statistics or pull request details are missing from the original dataset.

The project will focus only on metadata and repository activity information relevant to the research questions in order to keep the scope manageable.

Data Cleaning

The collected data will likely contain missing values, inconsistent formatting, duplicate records, and noisy text fields. Data cleaning steps may include:

- Removing duplicate pull requests
- Handling missing values
- Standardizing timestamps
- Cleaning pull request descriptions
- Normalizing AI tool names
- Filtering irrelevant records

Text-based fields may also be processed using simple natural language preprocessing methods such as:

- Lowercasing
- Tokenization
- Removing stop words
- Keyword matching

Exploratory Data Analysis

Exploratory Data Analysis (EDA) will be used to better understand patterns and trends in the dataset.

Possible visualizations include:

- Bar charts showing AI coding agent frequency
- Merge rate comparisons
- Histograms for review duration
- Boxplots for additions/deletions
- Correlation heatmaps
- Scatter plots comparing metadata features
- Time-based trends in AI-assisted pull requests

Visualization libraries may include:

- Matplotlib
- Seaborn

The EDA phase will help identify useful features, trends, and possible outliers before moving to predictive analysis.

Comparative Analysis

The project will compare pull request outcomes between different AI coding agents.

The comparisons may include:

- Merge success rates
- Average review duration
- Number of review comments
- Pull request size
- Repository acceptance behavior
- Frequency of merged vs. closed pull requests

The analysis may also investigate whether certain repositories respond differently to AI-assisted pull requests.

Statistical Analysis

Simple statistical methods will be used to evaluate whether differences between AI coding agents are meaningful.

Possible methods include:

- Descriptive statistics
- Correlation analysis
- T-tests
- Chi-square tests
- Confidence intervals

The goal is not to make strong causal claims, but rather to identify patterns and relationships within the dataset.

Machine Learning / Prediction Models

The project will include a small predictive analytics component.

Possible prediction goals include:

- Predicting whether a pull request will be merged
- Predicting whether a pull request will receive extensive review comments
- Predicting review duration categories

Possible machine learning models include:

- Logistic Regression
- Decision Trees
- Random Forest

Input features may include:

- Pull request size
- Number of commits

- Pull request text length
- Repository activity
- Review comments
- Labels/tags
- AI coding agent mentions

The project will compare model performance and identify which features contribute most strongly to predictions.

Evaluation Metrics

The machine learning models may be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

If time-based prediction tasks are attempted, possible evaluation metrics include:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

The project will also evaluate feature importance where possible.

Planned Workflow

The planned workflow for the project is:

1. Access and load the AIDev dataset
2. Explore dataset structure and metadata
3. Filter and preprocess relevant pull request data
4. Clean and standardize metadata fields
5. Perform exploratory data analysis and visualization
6. Compare pull request outcomes between AI coding agents
7. Apply statistical analysis methods
8. Train and evaluate prediction models
9. Analyze limitations and threats to validity
10. Prepare the final report and presentation

Tools and Feasibility

The project mainly uses free and open-source tools. No major paid software or cloud infrastructure is expected to be required.

Planned tools include:

- Python
- pandas
- scikit-learn
- Jupyter Notebook
- Matplotlib
- Seaborn

Optional tools may include:

- Google Colab
- GitHub APIs

The team already has basic familiarity with Python, Jupyter Notebook, and common data analysis libraries. Since the AIDev dataset is already publicly available and accessible, the project is considered feasible within the semester timeline.

Potential Limitations / Threats to Validity

One important limitation is that identifying truly AI-generated or AI-assisted pull requests is difficult. Some developers may use AI tools without mentioning them in pull request descriptions or commit messages. Because of this, some AI-assisted contributions may not be fully captured in the dataset.

Another limitation is that repository behavior may vary significantly between projects. Some repositories may have stricter review processes or different contribution standards, which can affect pull request outcomes independently of the AI tool being used.

Additional threats to validity include:

- Missing or incomplete metadata
- Sampling bias
- Differences in repository size and activity
- Inconsistent naming of AI tools
- API limitations
- Correlation not implying causation

The project findings will therefore be treated as exploratory rather than fully causal conclusions.

Expected Results

The project is expected to produce both descriptive and predictive insights related to AI-assisted software development.

Expected outcomes include:

- A comparison of AI coding agent usage frequency
- Analysis of pull request merge behavior
- Visualizations showing repository and review trends
- Identification of metadata features linked to successful pull requests
- Basic prediction models for merge outcomes
- Discussion of dataset limitations and biases

The project may show that certain metadata features are associated with higher merge success or faster reviews. However, the study also expects that repository-specific behavior and contributor experience may influence results alongside AI tool usage.

Possible future extensions could include:

- Sentiment analysis of review comments
- Code quality analysis
- Long-term trends in AI adoption
- NLP-based analysis of pull request text
- Security-related analysis of AI-generated code

Task Allocation and Timeline

Week	Planned Task	Team Member(s)
Week 1	Literature review and dataset familiarization	_____
Week 2	Dataset exploration and preprocessing	_____
Week 3	Data cleaning and feature extraction	_____
Week 4	Exploratory data analysis and visualization	_____
Week 5	Comparative and statistical analysis	_____
Week 6	Machine learning model development	_____
Week 7	Model evaluation and interpretation	_____
Week 8	Writing results and discussion sections	_____
Week 9	Final report preparation	_____
Week 10	Video presentation and final revisions	_____

The team plans to coordinate progress using shared notebooks, GitHub repositories, and regular meetings throughout the semester.

Core References

1. [AIDev Dataset \(MSR Challenge\)](#)
2. [GitHub REST API Documentation](#)
3. [GitHub GraphQL API Documentation](#)
4. [scikit-learn Documentation](#)
5. [pandas Documentation](#)
6. Kalliamvakou, E. et al. “The Promises and Perils of Mining GitHub Data.”
7. Bird, C. et al. “Mining GitHub: Challenges and Opportunities.”
8. Menzies, T., Williams, L., & Zimmermann, T. “Perspectives on Data Science for Software Engineering.”
9. Recent research papers from:
 - [IEEE Xplore](#)
 - [ACM Digital Library](#)
 - [arXiv](#)